ENTERPRISE AI INFRASTRUCTURE • OPEN SOURCE PLATFORM

BIFROST GATEWAY

One Gateway. Every AI. Infinite Possibilities.

Unified Multi-Provider API | Intelligent Model Routing | pgvector Semantic Cache | 18ms Failover | MCP Tool Gateway

< 4ms	71%	15+	99.99%
Internal Latency	Cost Reduction	Providers Connected	Uptime SLA

MARKET OPPORTUNITY

The Enterprise AI Infrastructure Crisis

Why direct multi-provider API integration creates operational chaos, outages, and runaway costs.

Vendor Lock-in & Sprawl

CRITICAL FRICTION

- Teams separately integrate OpenAI, Anthropic, Gemini, DeepSeek, and Bedrock.
- Inconsistent SDKs, divergent error codes, and unique streaming transcoders.
- Engineering time wasted rewriting client code whenever new models release.
- Zero single-pane-of-glass visibility across enterprise LLM traffic.

Unreliable Upstream Outages

SLA BREACHES

- Rate limits (HTTP 429) and provider outages (503/500) crash production apps.
- Developers build brittle in-app retries that cause destructive retry storms.
- Lack of dynamic failover between cloud regions and alternative providers.
- No circuit breaker protection against runaway looping agents.

Runaway Token Spend

BUDGET WASTE

- Identical and repetitive queries repeatedly billed at full frontier prices.
- Simple classification tasks routed to expensive flagship models.
- Absence of semantic caching and per-tenant budget kill-switches.
- API keys hardcoded across microservices with zero access control.

PRODUCT VISION

The Bifrost Gateway Solution

One unified high-throughput infrastructure platform connecting applications with all AI providers.

Unified Drop-in OpenAI API

STANDARDIZED CONTRACT

- Developers point any OpenAI client to Bifrost by updating base_url.
- Zero code rewrite required across Python, TypeScript, Java, and Go.
- Standardized REST, SSE streaming, and multimodal payload encoding.

Intelligent Dynamic Router

SUB-3MS DECISION

- Evaluates prompt intent, context window, latency SLAs, and cost ceilings.
- Automatically routes each request to the optimal model in < 3ms.
- Weighted load balancing and tag-based policy distribution.

pgvector Semantic Caching

ZERO-COST INFERENCE

- Hybrid exact match and cosine similarity caching using PostgreSQL & Redis.
- Serves repeated or semantically equivalent queries in < 4ms at $0 cost.
- Slashes enterprise monthly token bills by 35% to 72%.

18ms Disaster Recovery Failover

100% 200 OK DELIVERY

- Integrated circuit breakers detect 429 rate limits or upstream 503 outages.
- Instantly falls back to secondary frontier models with 0 app downtime.
- Enterprise Virtual Keys with monthly spend caps and PII redaction.

ENGINEERING DESIGN

Complete 6-Layer System Architecture

High-throughput Go data plane separated from the administrative control plane.

Layer 01: Client & Ingress

Web/Mobile Apps, Python/TS SDKs, Autonomous Agents, Cursor, LangChain

Layer 02: Gateway Edge & Security

Go FastHTTP reverse proxy, Token-bucket rate limiting, PII/PCI redaction

Layer 03: Intelligence & Routing

pgvector semantic distance lookup, Multi-objective router, Circuit breakers

Layer 04: Provider Adapters

OpenAI, Anthropic, Gemini, DeepSeek, Groq, Bedrock, Azure, Mistral, Ollama

Layer 05: Distributed Data & Cache

PostgreSQL 16 pgvector, Redis 7 cluster for rate limits, AES-256 key vault

Layer 06: Operations & Governance

OpenTelemetry distributed W3C tracing, Prometheus /metrics, 14-screen Console

ECOSYSTEM COMPATIBILITY

Universal Multi-Provider Coverage

Seamlessly interconnect 15+ commercial frontier, open-weight, and on-premise inference engines.

OpenAI

GPT-4o, GPT-4o-mini, o1, o3-mini

Latency: 142ms | Pricing: $2.50 / $10.00

Anthropic Claude

Claude 3.5 Sonnet, 3.5 Haiku, Opus

Latency: 168ms | Pricing: $3.00 / $15.00

Google Gemini

Gemini 1.5 Pro (2M Context), 2.0 Flash

Latency: 118ms | Pricing: $1.25 / $5.00

DeepSeek

DeepSeek R1 (Reasoning), DeepSeek V3

Latency: 95ms | Pricing: $0.14 / $0.28

Groq LPU

Llama 3.3 70B (480 tokens/sec)

Latency: 38ms | Pricing: $0.59 / $0.79

AWS Bedrock

Claude 3.5, Titan, Command R+

Latency: 185ms | Pricing: Enterprise IAM

Azure OpenAI

GPT-4o Enterprise Private Endpoint

Latency: 154ms | Pricing: HIPAA BAA

Mistral AI

Mistral Large 2, Codestral 2501

Latency: 122ms | Pricing: EU Sovereign

Ollama Local

Llama 3.2, Qwen 2.5 (Air-Gapped)

Latency: 62ms | Pricing: $0.00 (Self-Hosted)

TRAFFIC OPTIMIZATION

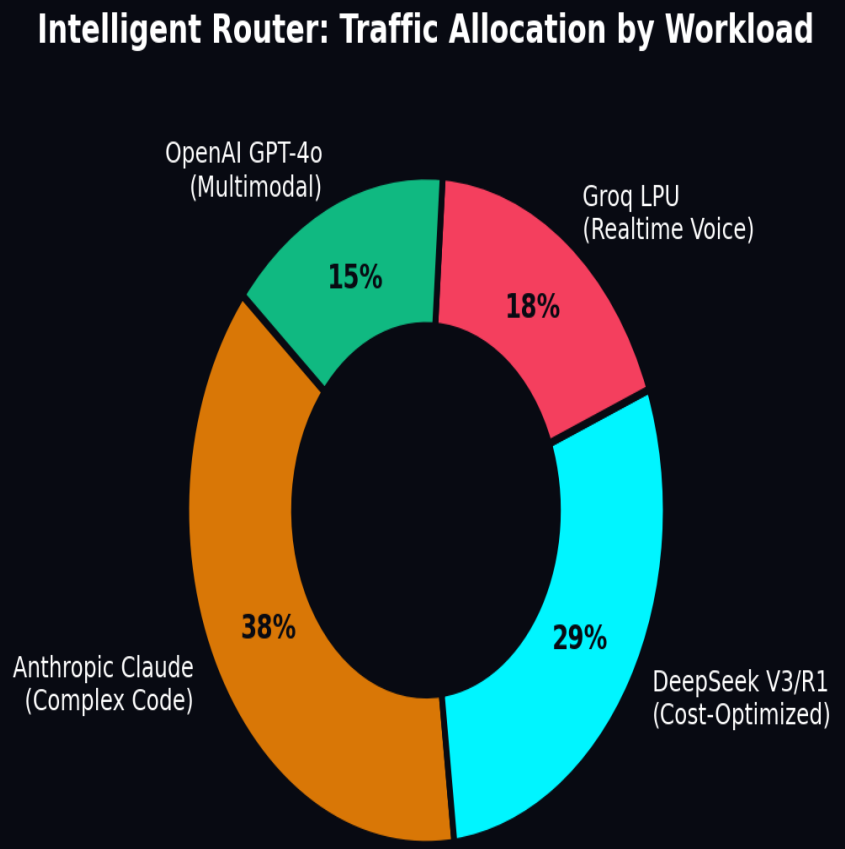
Intelligent Model Routing Engine

Dynamic, multi-objective optimization routes requests to the ideal model in under 3 milliseconds.

Routing Policies & Strategies

POLICY RULES

- Cost-Optimized Strategy: Enforces max $0.50/1M constraint, routing routine queries to DeepSeek V3 or Gemini Flash (-78% spend).
- Sub-50ms Latency Strategy: Routes conversational voice workloads to Groq LPU (38ms P99) for immediate speech output.
- High-Context Coding Strategy: Automatically sends complex AST synthesis and refactoring to Claude 3.5 Sonnet.
- Long-Form Document Strategy: Utilizes Gemini 1.5 Pro's native 2M token context window without brittle chunking.
- Weighted Enterprise Balance: Splits volume across Azure OpenAI (50%), Bedrock (30%), and Google Cloud (20%).



HIGH AVAILABILITY

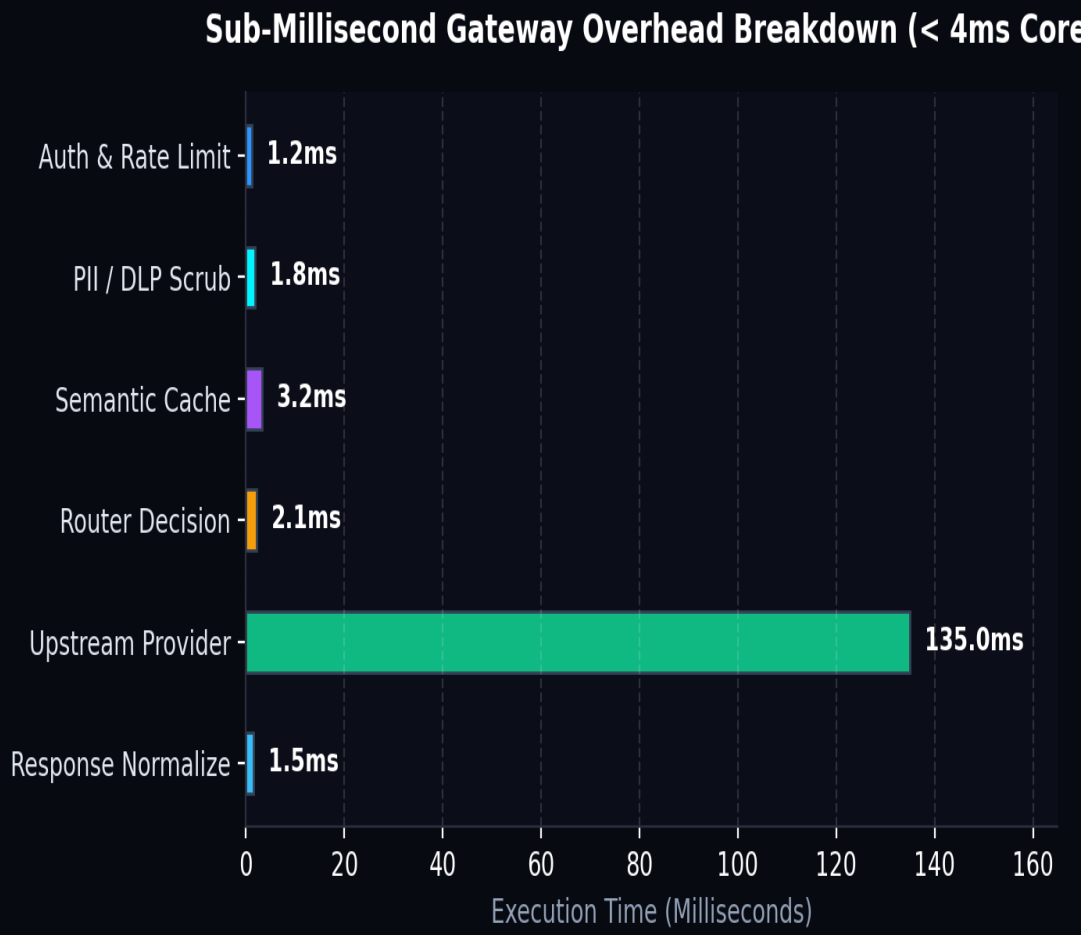
Zero-Downtime Automatic Failover (18ms)

Circuit breakers and bounded fallback chains eliminate 99.9% of upstream AI downtime.

Disaster Recovery Invariants

CIRCUIT BREAKERS

- Upstream Outage Interception: Catches HTTP 429 (Rate Limits) and 503/500 errors before applications notice.
- 18ms Instant Fallback: Cascades to alternative provider (e.g. OpenAI -> Claude 3.5 -> Azure OpenAI) in under 20ms.
- Circuit Breaker State Machine: Trips to OPEN state after 3 consecutive failures with 30s cooldown timer.
- Bounded Retry Budget: Clamps retries to prevent catastrophic downstream retry storms.
- 100% Client Availability: Returns HTTP 200 OK without surfacing provider downtime to users.



ECONOMIC EFFICIENCY

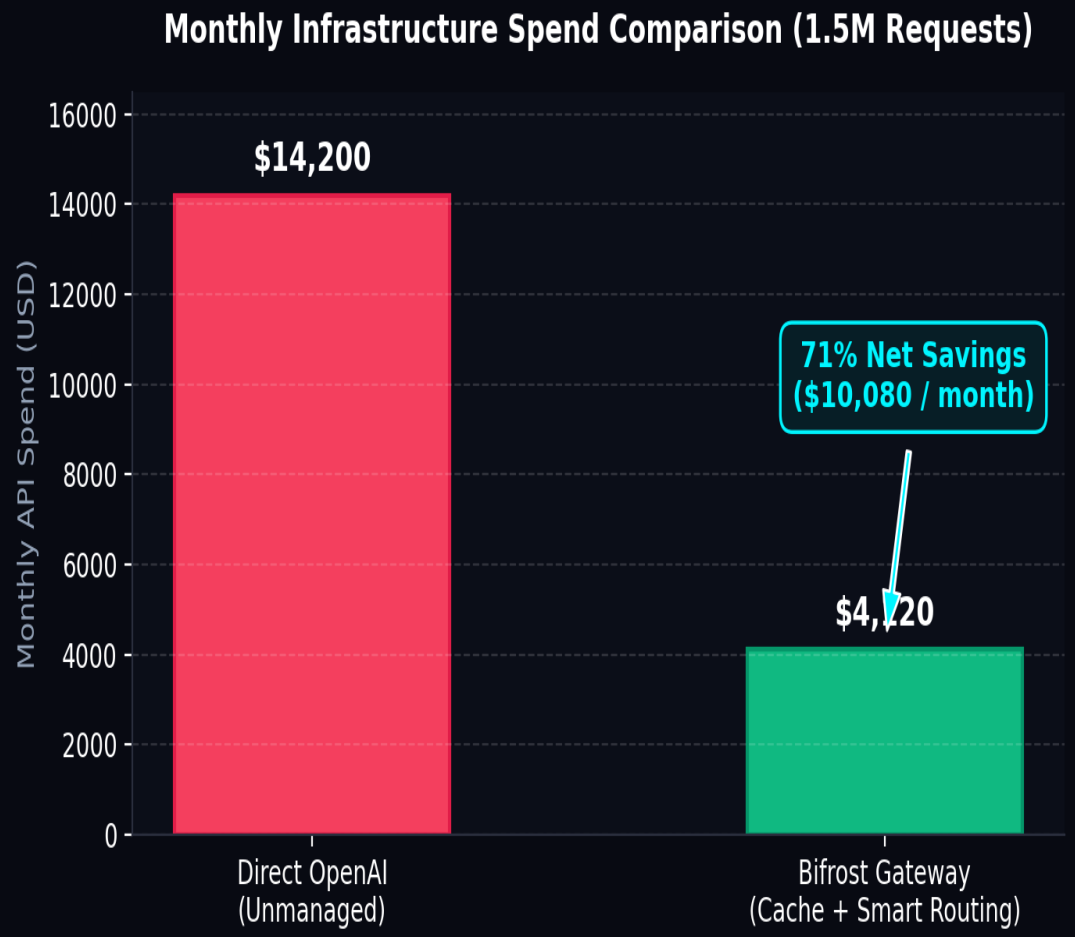
Cost Intelligence & Semantic Vector Caching

Cut enterprise AI API expenditures by 71% through intelligent memory and model tiered routing.

Semantic Cache Architecture

VECTOR MEMORY

- Cosine Distance Lookup: Embeds queries and queries pgvector IVFFlat index in under 4ms.
- Configurable Similarity Threshold: Dial between 0.80 (loose high-hit) to 0.99 (strict near-exact) precision.
- Zero-Cost Responses: 35% of repetitive queries answered with 0 upstream tokens billed.
- TTL & Invalidation: Automatic time-to-live policies and tenant-isolated namespace boundaries.
- Annualized ROI: Saves $120,000+ per year on high-volume 1.5M requests/month workloads.



PERFORMANCE BENCHMARKS

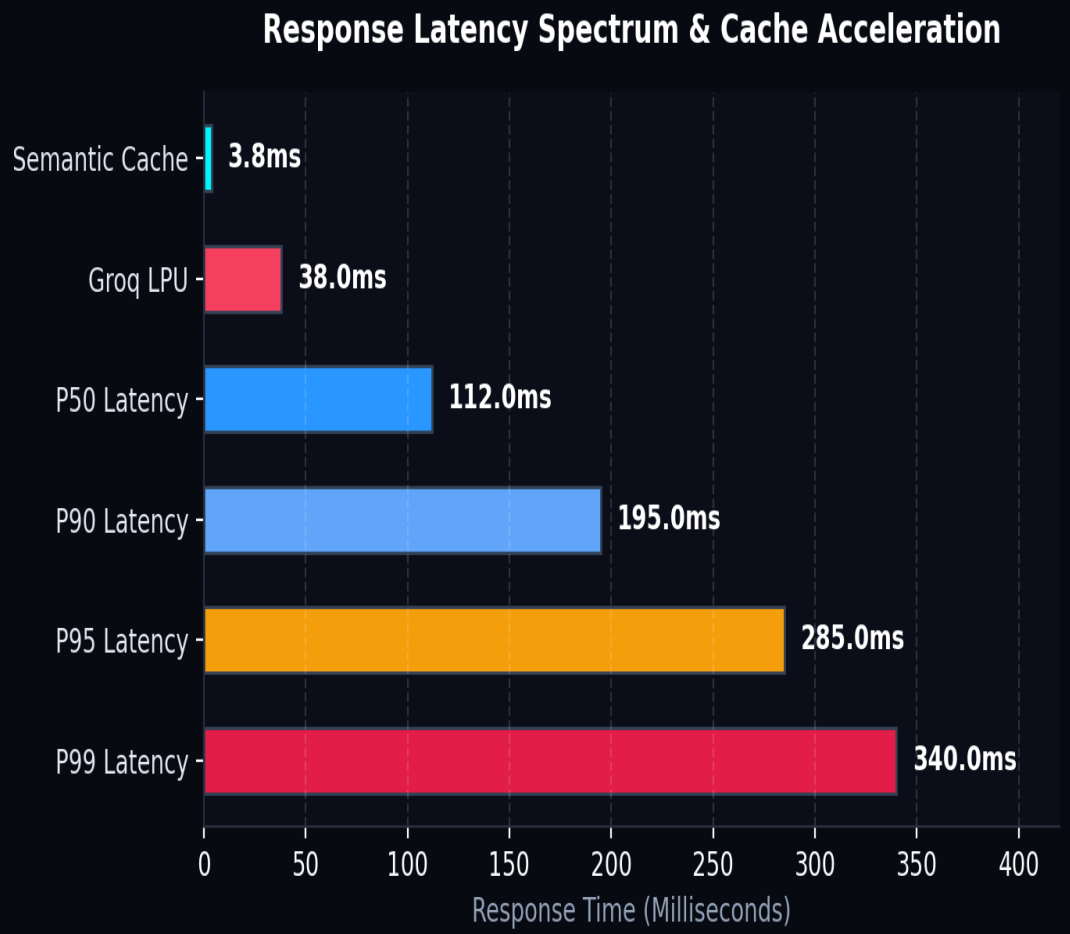
Latency Spectrum & Real-Time Performance

Sub-4ms internal gateway overhead with hardware-accelerated LPU streaming.

Benchmark Characteristics

SPEED METRICS

- Internal Overhead: < 4ms added latency for full authentication, caching, guardrails, and routing.
- Semantic Cache Speed: 3.8ms total roundtrip response for vector cache hits.
- Groq LPU Integration: 38ms time-to-first-token with 480 tokens/sec streaming throughput.
- P95 / P99 SLA: P50 latency 112ms, P99 guaranteed under 340ms across multi-cloud regions.
- SSE Streaming Transcoder: Converts divergent upstream streams to standardized SSE in < 1.2ms.



DATA PROTECTION

Enterprise Security, Virtual Keys & Guardrails

Never expose root provider secrets. Issue scoped virtual keys with automated budget kill-switches.

Tenant Virtual API Keys

TOKEN SCOPING

- Issue isolated virtual keys to specific teams, microservices, or frontend clients.
- Clamp monthly budgets (e.g. $2,500/mo) with automated reject upon ceiling.
- Restrict keys to specific allowed model whitelists and RPM rate limits.

Real-time DLP & PII Redaction

PRIVACY SHIELD

- Sanitize credit card PANs, SSNs, and patient identifiers before packets leave VPC.
- Full HIPAA, GDPR, and PCI-DSS compliance conformance ready.
- Custom regex and embedding-based sensitive entity maskers.

Prompt Injection Defense

THREAT MITIGATION

- Block adversarial jailbreak attempts ('Ignore all previous instructions...').
- Detect toxic outputs and hate speech with real-time classification gates.
- Cost velocity spike breaker ($50/min cap) prevents infinite agent loops.

SOC2 Immutable Audit Trails

AUDIT COMPLIANCE

- Every request, key rotation, policy change, and PII redaction event logged.
- Full W3C trace context propagation across every microservice hop.
- Export audit records to Splunk, Datadog, or AWS S3 cold storage.

AUTONOMOUS AGENTS

Model Context Protocol (MCP) Tool Bridge

Governed tool execution and human-in-the-loop approval gates for agentic workflows.

Agentic Integrations

AGENT CLIENTS

- Direct bridge for LangChain, CrewAI, AutoGen, Cursor, and Claude Code.
- Agents connect to Bifrost via standardized OpenAI tool-calling semantics.
- Eliminates unmonitored raw shell or direct production database access.
- Detailed tool call latency, token usage, and error metrics recorded.

Registered MCP Tools

ENTERPRISE TOOLS

- PostgreSQL Enterprise MCP: Read-only parameterized queries with limit clamps.
- GitHub Org MCP: Secure PR creation and issue tracking with branch protection.
- Slack Alerts MCP: Incident escalation dispatch to #ai-ops channels.
- Brave Web Search MCP: Real-time web index search with content extraction.

Approval Gates & Governance

HUMAN GATES

- Granular Execution Policies: Toggle tools between 'Auto-Approved' vs 'Approval Required'.
- Human-in-the-Loop: High-risk tools (e.g. create PR, send alert) require explicit admin sign-off.
- Parameter Validation: JSON schema enforcement blocks SQL injection attempts.
- Complete Audit Playback: Replay agent execution steps for incident forensics.

PRODUCTION READY

Deployment, Administration & Open-Source Roadmap

Enterprise self-hosting with Docker, Kubernetes, and 14-screen admin console.

Self-Hosted Deployment Stack

TURNKEY DEPLOYMENT

- Docker Compose: Single command bootstrap with Redis and pgvector.
- Kubernetes & Helm: Multi-replica deployment, mTLS ingress, and HPA auto-scaling.
- 22MB Static Go Binary: Zero external runtime dependencies for maximum security.

14-Screen Admin Console

CONTROL PLANE

- Built-in Next.js 15 dark-mode control plane at /dashboard.
- Live waterfall trace inspector with sub-millisecond execution spans.
- Configurable semantic similarity thresholds and one-click cache purges.

Proven Economic ROI

EXECUTIVE ROI

- 71% reduction in monthly API spend ($10,080/mo saved on 1.5M requests).
- Zero vendor lock-in allows negotiating volume discounts across providers.
- SLA protection prevents catastrophic customer churn during provider outages.

Open Source & Next Steps

GET STARTED TODAY

- Released under Apache 2.0 open-source license on GitHub.
- Explore live interactive playground at http://localhost:3000/playground.
- GitHub Repository: https://github.com/Ranjeet7680/BIFROST-GATEWAY.